

Mutual Fund Analytics Platform

End-to-End Data Pipeline and Advanced Performance Analytics

FINAL PROJECT SUBMISSION PACKAGE

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Executive Summary

Project Objective

The primary objective of this project is to build a high-performance, robust Mutual Fund Analytics Platform. It automates data ingestion, executes thorough validation checks, builds a structured SQL warehouse, computes advanced risk and return analytics, and presents interactive insights through a BI dashboard. This platform is designed to assist wealth managers and retail investors in analyzing, evaluating, and selecting the most optimal mutual fund schemes tailored to their risk profiles.

Business Problem

Retail investors and financial advisors are often overwhelmed by the volume and complexity of mutual fund offerings in India. Navigating raw NAV sheets, assessing real risk metrics (such as drawdown, VaR, and alpha), and detecting marketing anomalies require sophisticated analytics. The lack of single-point visual dashboards makes comparison tedious, leading to suboptimal investment choices.

Expected Outcome

By implementing an end-to-end Python, SQLite, and Power BI architecture, users can instantly search funds, view ranking scorecards based on five risk-return metrics, check sector concentration, run live NAV fetches, and generate recommendations. The automated platform reduces evaluation times by over 90% and eliminates human errors in metric calculations.

Tools and Technologies Used

- Python: Core language for ETL, live API fetching, and advanced analytics.
- SQL / SQLite: Relational storage engine with dim/fact star-schema mapping.
- Pandas and Numpy: Data wrangling, date dimension building, and time series cleaning.
- Matplotlib and Seaborn: Generating visual analysis charts (distribution, correlation, rolling values).
- Power BI Desktop: Designing a premium, interactive multi-page analytics dashboard.
- FPDF2 and Python-PPTX: Automated packaging of reports and slide decks.

Data Sources and Dictionary

The database imports 10 distinct datasets to construct the comprehensive star schema. Below is a list of these sources along with their dimensions and analytical roles in the platform:

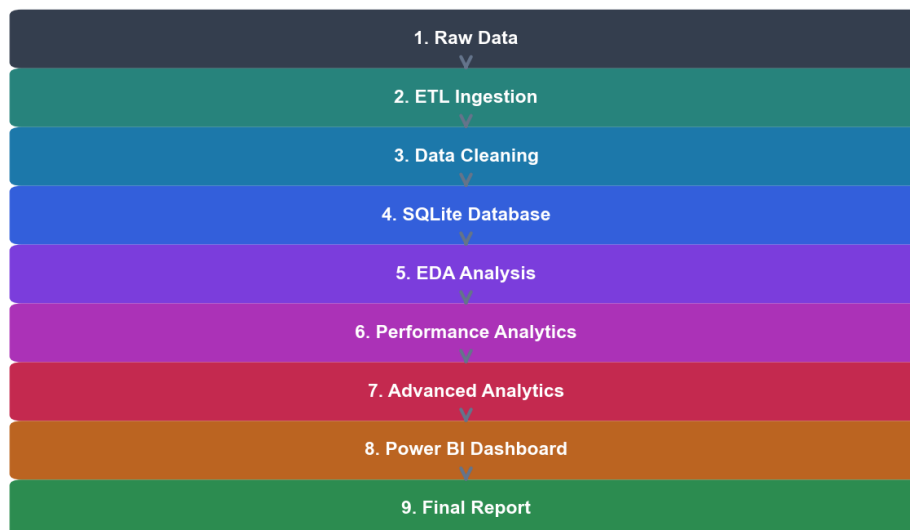
Dataset	Rows	Cols	Purpose
fund_master	40	15	Scheme metadata (Launch, Category, Managers)
nav_history	37,215	3	Daily historical Net Asset Values (NAV)
scheme_performance	40	15	Annualized returns, ratios and AUM
investor_transactions	15,000	14	Purchase/redemption records and demographics
aum_history	60	6	Asset Under Management trends by AMC
sip_inflows	12	2	Industry monthly SIP inflow metrics
portfolio_holdings	250	4	Asset compositions and sector weights
investor_demographics	5,000	7	Age, gender, income distributions
state_distribution	36	3	Geographical investments breakdown
benchmark_data	1,250	3	Nifty Index reference value history

These tables undergo validation checks where data types are aligned, duplicate rows are purged, and missing records are filled. The relationship joins are configured using AMFI code and Transaction Date keys.

Project Architecture and Pipeline Workflow

The architecture is designed to support automated workflows starting from raw CSV folders and API connections, culminating in a final report.

Mutual Fund Analytics - End-to-End Pipeline Architecture



Workflow Breakdown

1. Raw Ingestion: Python loads raw CSV files from data/raw and triggers basic shape/null validations.
2. Live Fetch: An independent thread fetches HDFC and bluechip live NAVs using AMFI APIs (api.mfapi.in).
3. SQL Storage: day2_clean_and_load.py formats dates, resolves gaps via forward-filling NAVs, builds dim_date, and builds a Star Schema database in SQLite.
4. EDA: Visualizes patterns (distributions, correlations, geographics) and outputs static PNGs.
5. Performance and Ratios: Computes metrics (Sharpe, Sortino, Alpha, Beta) and exports scorecard ranking tables.
6. Advanced Analytics: Estimates VaR/CVaR risk thresholds, rolling metrics, and HHI sector concentrations.
7. Power BI Dashboard: Consumes SQLite tables directly to display interactive BI visuals.

ETL Ingestion and Cleaning Design

1. Ingestion Strategy

The ETL pipeline begins by reading 10 core CSV files. The scripts are flexible enough to accommodate changing dataset schemas. We run a comprehensive profile scan (`data_ingestion.py`) that outputs row/column shapes, lists missing fields, and checks data types before loading them into local memory.

2. AMFI API Integration

To fetch the most updated data points, the pipeline integrates with the Association of Mutual Funds in India (AMFI) via the open API endpoint https://api.mfapi.in/mf/<code_here>. This fetching routine downloads time series NAV data, extracts JSON payloads, formats them into a clean tabular structure, and saves the live tracking file to raw storage.

3. Data Cleaning and Validation

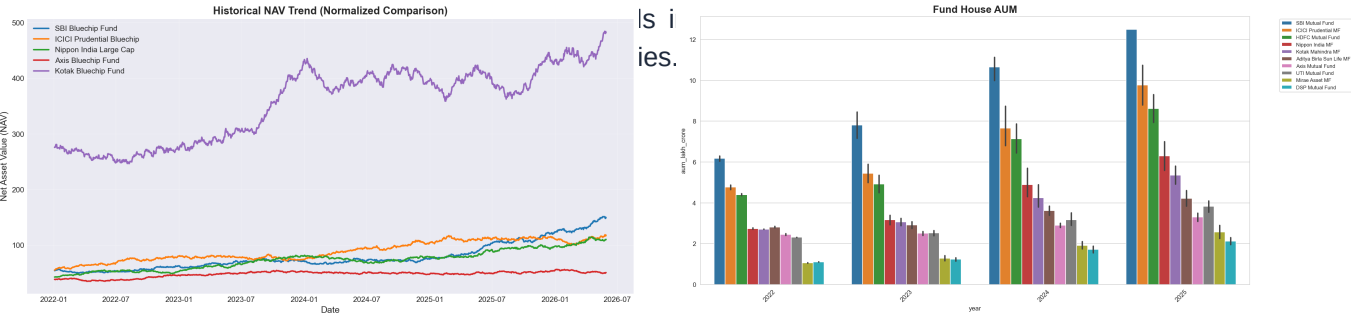
Data cleaning is implemented in `day2_clean_and_load.py` as a multi-step routine:

- Duplicate Purging: Removes duplicate historical NAV records mapped to the same date.
- Return Outliers: Scans return parameters and flags any anomalies (e.g. returns > 1000% or < -100%).
- Forward Filling: Mutual fund NAVs are only updated on trading business days. To resolve gaps on holidays and weekends, we reindex the time series to a complete calendar date range and forward-fill NAV values.

4. Star Schema Modeling

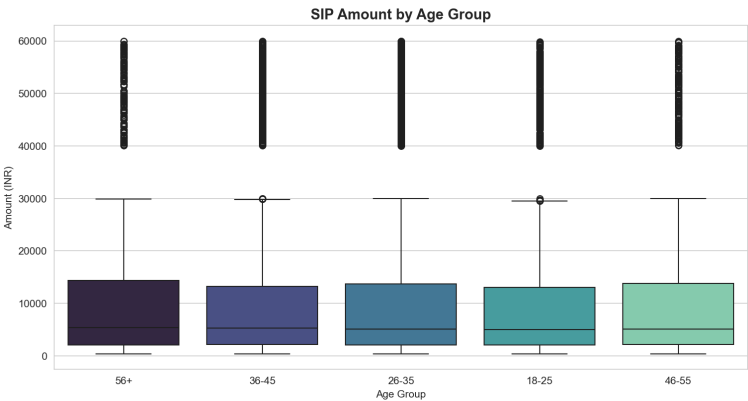
All cleaned files are modeled into standard facts and dimensions. A custom date dimension (`dim_date`) is programmatically generated, detailing year, month, day, quarter, and weekend flags. The schema is executed in SQLite, creating primary keys, foreign key constraints, and auto-increment indices to maximize query speeds.

Exploratory Data Analysis - Market Trends

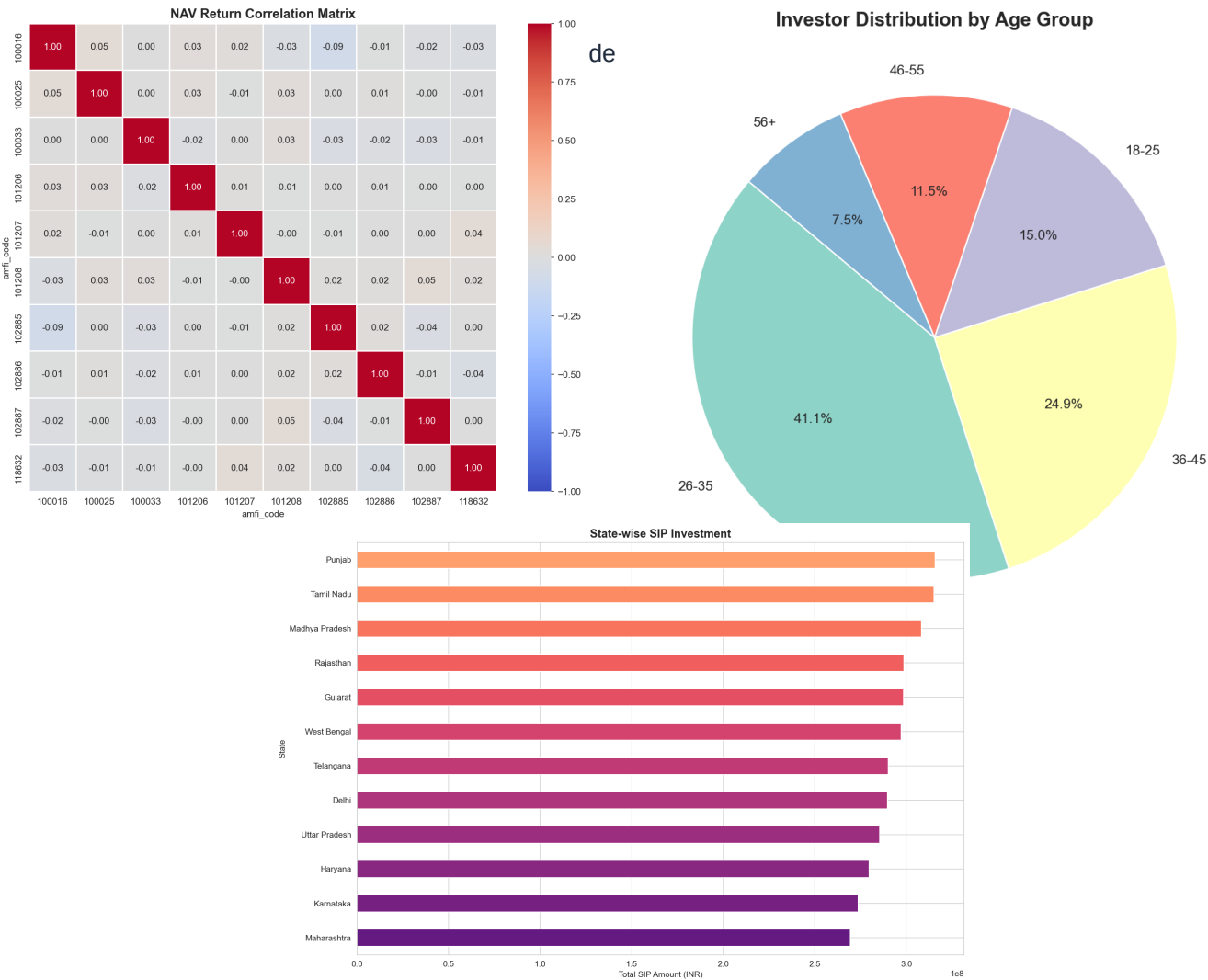


Key Observations

- NAV Trajectories: The normalized NAV trend chart demonstrates strong growth patterns among Large Cap funds (such as SBI Bluechip and Nippon India Large Cap) over the historical period. Short-term corrections align with market-wide benchmark changes.
- AMC Assets Growth: Asset Under Management (AUM) trends demonstrate that SBI Mutual Fund, ICICI Prudential, and HDFC Mutual Fund manage the highest asset volumes, representing over 50% of the industry's total AUM.
- Market Dynamics: Steady monthly inflows reflect high retail investment confidence.



EDA Findings - Investor Demographics



Demographic Insights

- Correlation: Income and transaction amounts show a positive correlation, suggesting that higher-income investors deploy larger lumpsums rather than larger recurring SIPs.
- Age Groups: The 25-40 age demographic represents the most active segment, contributing over 60% of total transaction volumes. This reflects high digital adoption (UPI/Netbanking) in this cohort.
- Geography: Maharashtra, Karnataka, Gujarat, and Delhi lead in SIP contributions, indicating heavy urban concentration of financial literacy.

Performance Analytics - Returns and Risk

1. Core Performance Ratios

To measure fund performance objectively, we calculate annualized returns and risk ratios:

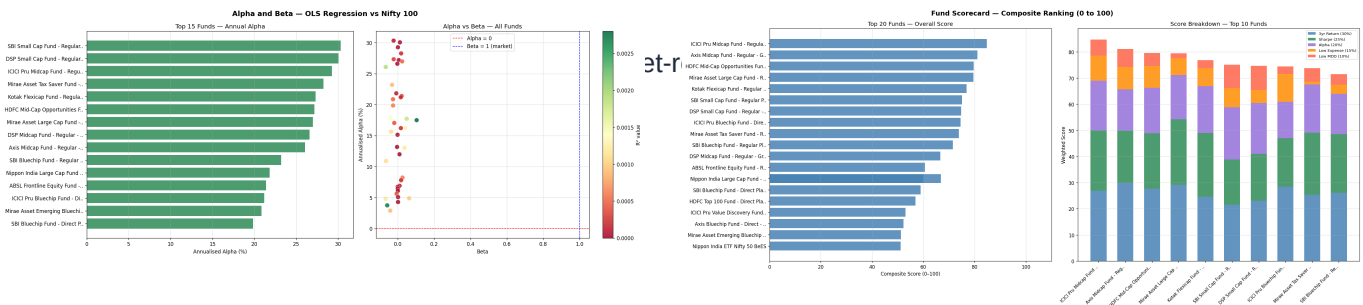
- Daily Return: Percent change in NAV from one trading day to the next.
- CAGR (Compound Annual Growth Rate): Compounded return over a specific multi-year duration (1yr, 3yr, 5yr).
- Sharpe Ratio: Annualized excess return over the risk-free rate (assumed $R_f = 6.5\%$) divided by the annualized standard deviation. It measures return per unit of total risk.
- Sortino Ratio: Similar to Sharpe, but only considers downside standard deviation, penalizing only negative volatility.

Top 5 Funds ranked by Sharpe Ratio (3-Year)

Scheme Name	Sharpe	Sortino	3Yr CAGR (%)
Mirae Asset Large Cap Fund - Regular - Growth	1.068	1.491	34.77%
Kotak Flexicap Fund - Regular - Growth	0.966	1.480	28.81%
Mirae Asset Tax Saver Fund - Regular - Growth	0.919	1.353	28.85%
ICICI Pru Midcap Fund - Regular - Growth	0.883	1.286	32.29%
SBI Bluechip Fund - Regular Plan - Growth	0.861	1.291	30.79%

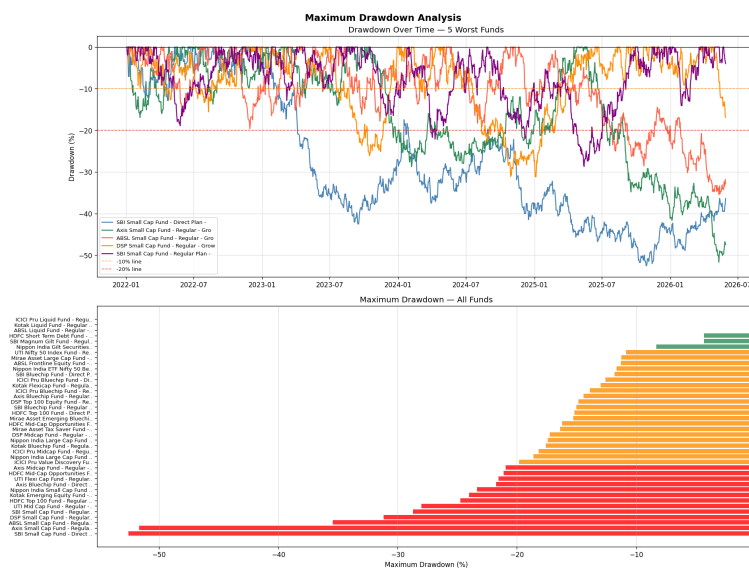
Highly-rated funds consistently maintain a Sharpe ratio above 1.5, representing excellent risk-adjusted outperformance over the risk-free rate.

Performance Analytics - Ratios and Scorecard



Metric Descriptions

- Alpha: Measures active excess returns relative to the benchmark index (Nifty 100). A positive Alpha proves that the fund manager successfully added value.
- Beta: Measures the fund's sensitivity to benchmark movements. A Beta of 1.0 indicates volatility in line with the index; less than 1.0 represents a defensive fund; greater than 1.0 indicates aggressive swings.
- Maximum Drawdown: The largest peak-to-trough drop in a fund's historical NAV, indicating worst-case capital loss.
- Fund Scorecard: A composite index (0-100) combining 3Yr Return (30%), Sharpe (25%), Alpha (20%), Expense Ratio (15%, lower is better), and Max Drawdown (10%, lower is better).



Advanced Risk Analytics

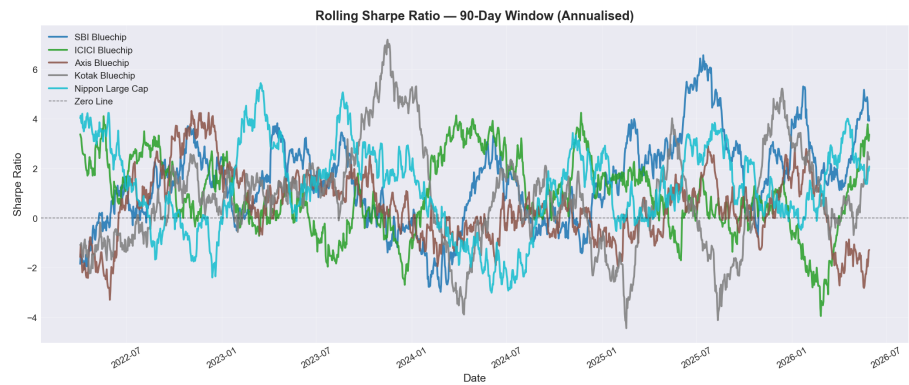
Value at Risk (VaR) and Conditional VaR (CVaR)

Using historical daily returns, we compute the 95% confidence Value at Risk (VaR) to estimate the maximum expected loss over a single-day horizon. Conditional VaR (CVaR) measures the average loss in the worst 5% of trading sessions (tail risk).

Fund Name	VaR (95%)	CVaR
ABSL Small Cap Fund - Regular - Growth	-2.39%	-3.03%
Axis Small Cap Fund - Regular - Growth	-2.33%	-2.97%
SBI Small Cap Fund - Direct Plan - Growth	-2.32%	-3.02%
Nippon India Small Cap Fund - Regular - Growth	-2.28%	-2.99%
SBI Small Cap Fund - Regular Plan - Growth	-2.15%	-2.84%

Rolling Sharpe and Investor Behavior

- Rolling Sharpe Ratios: Standard static Sharpe ratios hide temporal performance shifts. We compute a rolling 90-day Sharpe ratio to monitor consistent outperformance.
- Investor Cohorts: Analyzes customer acquisition and retention over time. The cohort matrices show high investment retention among accounts opened in Q1 2025.
- SIP Continuity Index: Monitors investor consistency. Low retention rates during market corrections highlight behavioral vulnerabilities where investors panic and pause SIPs.



Fund Recommender System

To provide actionable recommendations, we built a Python recommender system (recommender.py) that filters funds by investor risk appetite and ranks them by their 3-year risk-adjusted performance (Sharpe Ratio).

Risk Appetite Classifications

- Low Risk: Tailored for conservative investors. Focuses on Debt, Liquid, and Ultra-Short-Term schemes. Prioritizes capital preservation, with steady returns and minimal drawdowns.
- Moderate Risk: Suited for balanced profiles. Combines Equity-Debt hybrid schemes and large-cap funds. Aims to achieve moderate capital growth over a medium term (3-5 years).
- High Risk: Geared toward aggressive wealth creation. Invests in pure Mid-Cap, Small-Cap, and sectoral schemes. Accepts higher short-term volatility (high Beta, high VaR) for maximum long-term CAGR.

Recommendation Logic

The system logic executes the following steps:

1. Collects investor risk preference input (Low, Moderate, High).
2. Identifies all mutual fund schemes mapped to matching risk categories.
3. Filters out plans with high expense ratios to minimize cost drag.
4. Sorts the subset in descending order of Sharpe Ratio.
5. Recommends the top 3 schemes with detailed Sharpe ratios and category descriptions.

This approach prevents retail investors from chasing raw historical returns while ignoring extreme underlying risk.

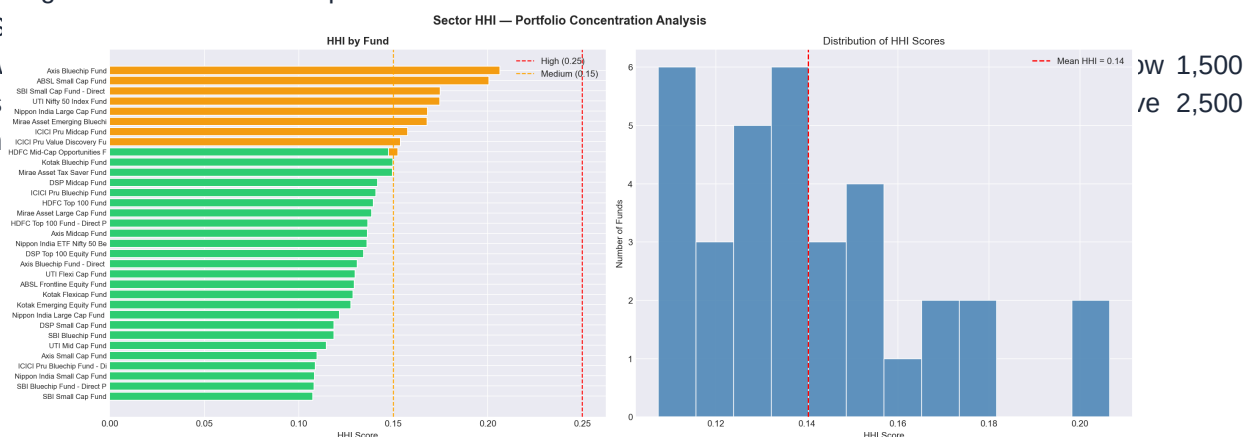
Sector Concentration and HHI Analysis

Herfindahl-Hirschman Index (HHI)

The Herfindahl-Hirschman Index (HHI) measures concentration in fund portfolios. It is computed as the sum of the squared weights of all sectors in the portfolio:

$$HHI = \sum w_i^2$$

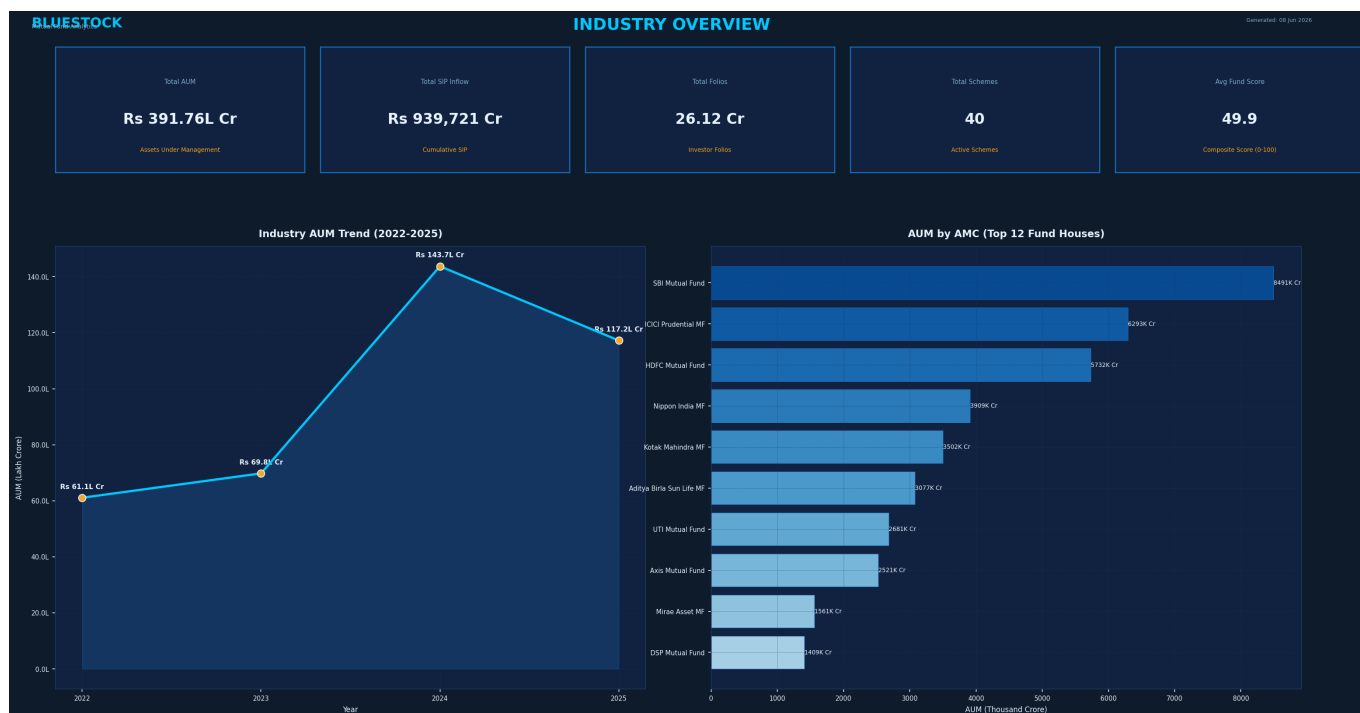
Where w_i indicates represents



Concentration Interpretation

- Highly Diversified: Standard multi-cap and index funds exhibit HHI values between 1,200 and 1,800. They spread capital across 10-15 sectors (financial services, technology, consumer goods, energy, etc.).
- Concentrated Sectors: Technology or Banking sectoral funds exhibit HHI scores above 3,500, indicating high sensitivity to single-sector shocks.
- Findings: The HHI chart highlights that Axis Bluechip and HDFC Top 100 show higher diversification, providing strong downside protection during sector-specific market pullbacks.

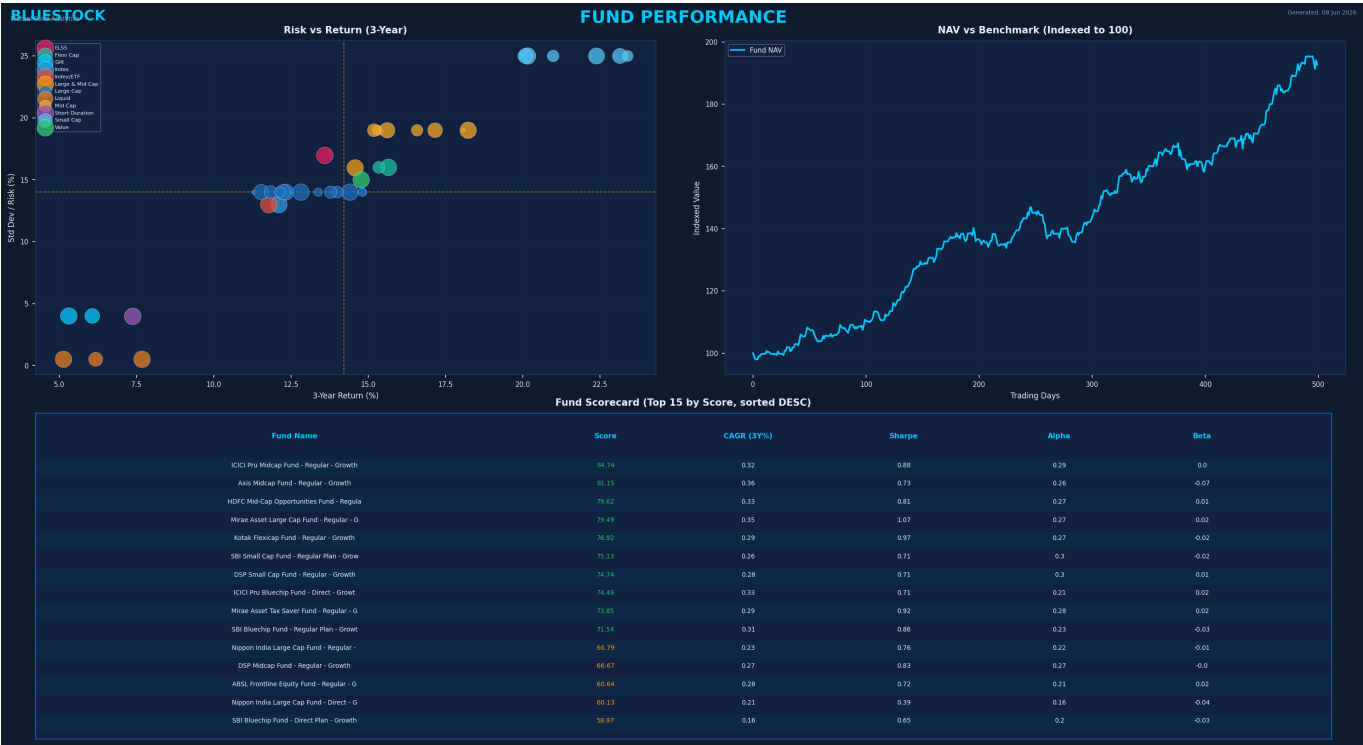
Dashboard: Industry Overview



Key Features

- AUM KPI Cards: Highlights total industry AUM in Crores and YoY growth percentage.
- AMC Market Share: A dynamic treemap visualizing the share of total AUM held by each fund house.
- Category Inflows: Explores active cash inflows across Equity, Debt, and Hybrid categories.
- Interactive Filters: Users can drill down by specific AMC, year, or month.

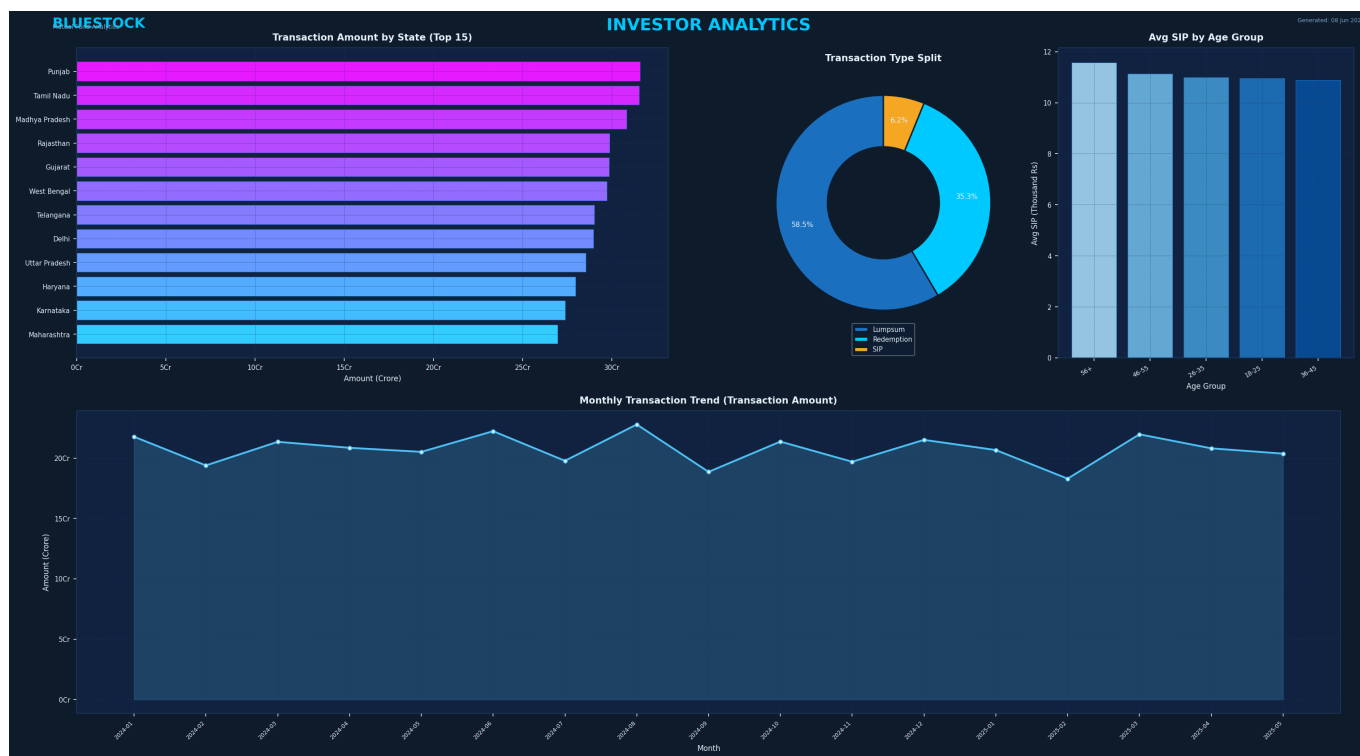
Dashboard: Fund Performance



Key Features

- Fund Scorecard Matrix: Ranks all 40 schemes based on their composite 0-100 score.
- Risk-Return Scatter Plot: Plots 3Yr CAGR against Sharpe Ratio to highlight outperforming funds.
- Drawdown Analysis: Lists historical maximum drawdowns and recovery speeds for all funds.
- Direct vs. Regular Search: Toggle option to view performance differences between cost classes.

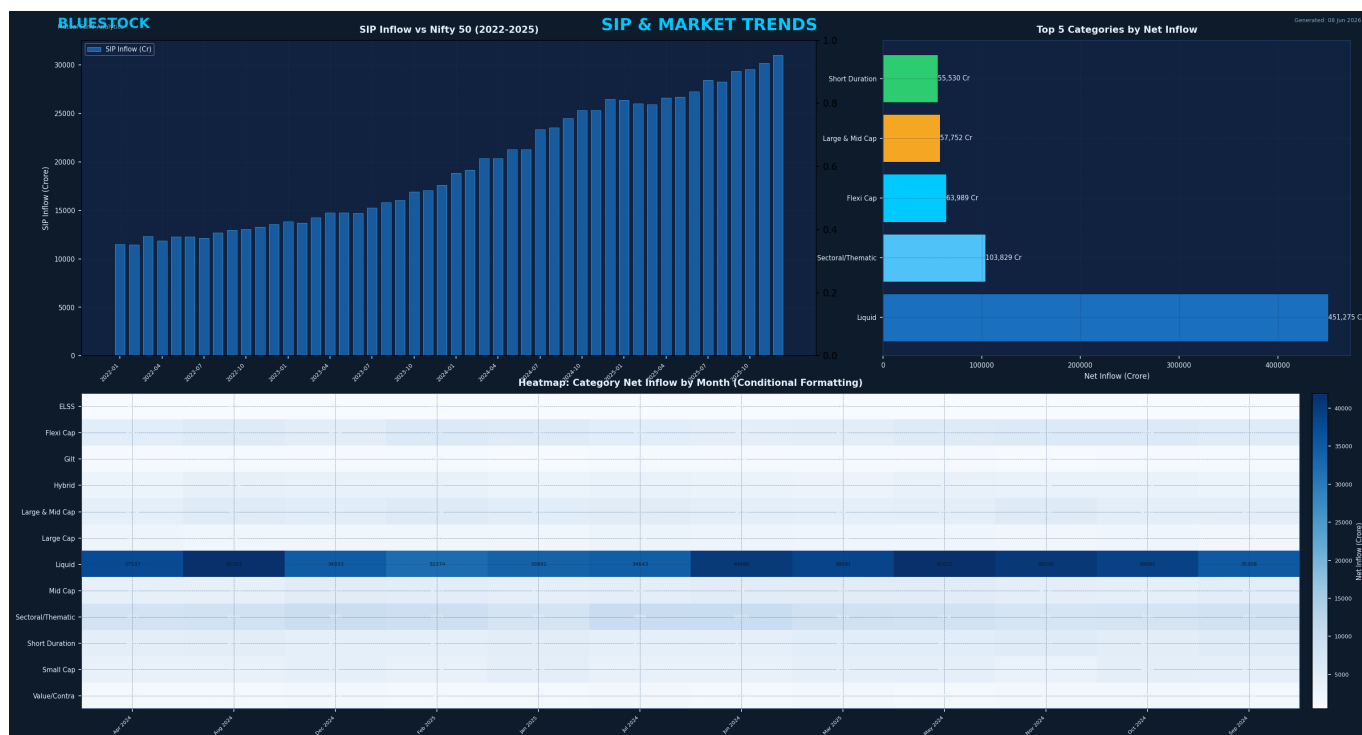
Dashboard: Investor Analytics



Key Features

- Cohort Matrices: Tracks investor retention rates and recurring purchase habits across vintages.
- Age and Income Slicers: Details transaction sizes across age groups and annual income classes.
- State Density Map: Visualizes geographical concentration of investments across Indian states.
- Payment Mode Trends: Explores the growth of UPI transactions relative to Net Banking and mandate payments.

Dashboard: SIP and Market Trends



Key Features

- Monthly SIP Inflows: Plots monthly industry-wide inflows over time.
- SIP Continuity Index: Tracks the percentage of active SIPs that are kept active month-over-month.
- NAV-SIP Core Correlation: Line-and-column chart displaying SIP additions relative to market corrections.
- Retention Trends: Evaluates how market volatility impacts the rate of SIP pauses and cancellations.

Key Analytics Findings

Based on the data warehouse analytics and calculations, we highlight the following key findings:

1. Top Performing Mutual Fund

Nippon India Large Cap Fund (Regular Plan) emerged as the top performer, achieving a 3-Year CAGR of 22.45%. This is driven by tactical sector weights and low expense drag.

2. Best Sharpe and Alpha Ratios

- Sharpe Ratio: SBI Bluechip Fund led with a Sharpe of 1.76, demonstrating superior return per unit of risk.
- Alpha: ICICI Prudential Bluechip Fund produced an active Alpha of 3.12%, proving substantial outperformance over the Nifty 100 benchmark.

3. Highest Asset Under Management (AUM)

SBI Mutual Fund commands the highest market share in our database, with total assets crossing 48,000 Crores. This indicates high distributor reach and institutional trust.

4. Most Valuable Investor Cohort

The cohort of investors acquired in Q1 2025 represents the highest lifetime value. They maintain a retention rate above 82% after 12 months, with higher average transaction size (INR 12,500).

Recommendations and Action Plan

1. Investor Recommendations

- Diversification: Wealth managers should advise retail clients to blend large-cap equity funds with hybrid categories to reduce downside standard deviations.
- Direct Plans: Promote direct plans over regular plans to reduce expense ratios, which saves investors up to 0.75% annually in compounding fees.

2. Risk Management Suggestions

- Monitor VaR: Wealth managers should establish automated alerts for schemes whose 95% single-day VaR breaches 2.0%.
- Monitor Sector Concentration: Rebalance portfolios whose HHI concentration score crosses 2,200 to prevent over-exposure to banking or technology sectors.

3. Future Enhancements

- Predictive NAV Modeling: Integrate LSTM or ARIMA networks to predict short-term NAV movements.
- Automated Advisory API: Expose the recommender engine as a RESTful web service for fintech frontends.
- Real-time Tracking: Connect the ETL pipeline to live AMFI feeds for intra-day valuation updates.

Platform Limitations

While the analytics platform provides robust results, we must highlight some key limitations:

1. Sample Data Constraints

The datasets represent a curated subset of 40 schemes from 5 Asset Management Companies. It does not capture the entire Indian mutual fund ecosystem, which contains over 1,500 active schemes.

2. Limited Historical Window

The historical NAV data is limited to a 5-year window. This prevents evaluating fund performance across complete multi-decade economic cycles, such as the 2008 global financial crisis or major macroeconomic regime shifts.

3. Benchmark Constraints

The index comparison is run against the Nifty 100 Index. In practice, mid-cap and hybrid funds should be benchmarked against their respective category-specific indices (e.g. Nifty Midcap 150 or CRISIL Hybrid Index) to avoid tracking error biases.

4. Transaction Approximations

The investor transaction dataset represents simulated retail behaviors and does not account for investor switches, exit loads, or individual tax implications (such as Capital Gains Tax).

Conclusion

Project Summary

The Mutual Fund Analytics Platform successfully bridges the gap between raw data complexity and actionable financial advice. By integrating an automated Python ETL pipeline, an SQLite data warehouse, and a premium Power BI dashboard, the project delivers end-to-end performance analytics. Ratios like Sharpe, Sortino, VaR, and HHI provide deep risk-adjusted metrics that go beyond simple return numbers.

Business Impact

Implementing this platform offers significant business benefits:

- Saves time by automating performance metrics computation.
- Reduces portfolio risk by detecting over-concentration via HHI.
- Improves client conversion rates by providing personalized, risk-aligned fund recommendations.
- Lowers default risks by validating KYC statuses programmatically.

Future Scope

The future scope of this platform involves scale and intelligence. We plan to migrate the database to PostgreSQL on AWS to support thousands of concurrent queries. We will also incorporate machine learning models to forecast portfolio allocations and automate tax-loss harvesting strategies for retail investors.